

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behaviour. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify it or its interactions; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access is robustly associated with green practice adoption (average marginal effect of 13 percentage points in

the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO2-emissions-monitoring outcome and with country fixed effects the primary sample’s single-wave design cannot support, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest’s feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, consistent with—though not proof of—SBFN-guided lending being channelled preferentially through the term instruments that actually finance long-horizon green capital expenditure. We interpret the combined evidence as showing that green banking policy’s binding constraint operates through firm-level absorptive capacity rather than through the country-level regulatory capacity its own institutional design targets.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model
JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner production. On paper, this represents one of the most ambitious diffusions of environmental financial regulation outside the OECD: economies as different in income, banking-sector depth, and political regime as Bangladesh, Morocco, Mongolia, and Indonesia now report, to the International Finance Corporation’s Sustainable Banking and Finance Network (SBFN), formal sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-risk disclosure rules for lenders. Whether that adoption changes what actually happens on a factory floor—whether a firm that can get a bank loan is any more likely to install a wastewater treatment system, switch to a more efficient boiler, or measure its own emissions—is a separate and considerably harder question, and it is the one this paper answers.

The premise that formal adoption of a policy is not the same as its operation in practice is, of course, not new to institutional economics. A large literature on state capacity and policy credibility has long argued that the

21 same law, regulation, or guideline can produce entirely different real-economy
 22 outcomes depending on the administrative and judicial infrastructure avail-
 23 able to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to
 24 green finance specifically, a parallel body of work has documented that green
 25 credit guidelines in China measurably reshape lending allocation and firm
 26 environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al.,
 27 2025), that firms’ green investment responds to the financing constraints they
 28 face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and
 29 that the World Bank Enterprise Surveys’ own Green Economy module is a
 30 workable lens on firm-level environmental behaviour across dozens of coun-
 31 tries at once (Asif et al., 2025; Phan, 2026). What is missing is the connection
 32 between these two literatures: a direct test, at firm level and across a wide
 33 cross-section of the very economies the green-banking-policy movement tar-
 34 gets, of whether the credit-to-green-investment channel that these guidelines
 35 are designed to activate is actually stronger where the guidelines exist—and
 36 whether that depends on the state capacity to make the guideline credible in
 37 the first place.

38 This paper closes that gap using firm-level data from two independently
 39 assembled samples of World Bank Enterprise Survey (WBES) waves. The
 40 primary sample covers 47 country-years fielded between 2018 and 2023 that
 41 carry the Bank’s full Green Economy module—a battery of questions on emis-
 42 sions monitoring, energy audits, and the adoption of climate-friendly produc-
 43 tion measures—alongside the Surveys’ standard access-to-finance module; a
 44 second, global sample extends this to 162 further country-years fielded be-
 45 tween 2021 and 2026, reaching for the first time in this literature into high-

46 income economies with no SBFN involvement at all (Section 4). We match
 47 each economy to its Sustainable Banking and Finance Network policy status
 48 as of its survey year, distinguishing genuine ex-ante policy adopters from
 49 economies that joined only afterward, and to the World Bank’s Worldwide
 50 Governance Indicators as a measure of the regulatory state capacity avail-
 51 able to make that policy credible. Econometrically, we depart deliberately
 52 from the pooled fixed-effects specification that has become the default in
 53 this literature. Because SBFN status is fixed within country in a single-wave
 54 cross-section, a country fixed effect absorbs it entirely, and a naive interaction
 55 term estimated by pooled logit—which we report first, as a baseline, precisely
 56 to make this limitation visible rather than to hide it—returns no detectable
 57 policy-moderation effect at all. We show that this null result is not evidence
 58 of an absent relationship so much as a symptom of forcing a fundamentally
 59 hierarchical question (does a level-2, country-level policy variable reshape
 60 a level-1, firm-level financing relationship?) into a flat, single-level model
 61 with too few country clusters to lean on for inference (Cameron and Miller,
 62 2015). We then re-estimate the same relationship with tools built for exactly
 63 this structure: a Bayesian hierarchical logistic model with country-varying
 64 slopes (Gelman and Hill, 2007; Gelman et al., 2020), a causal-forest / double-
 65 machine-learning (DML) estimator (Wager and Athey, 2018; Chernozhukov
 66 et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that lets the treat-
 67 ment effect of credit access vary flexibly across the institutional landscape
 68 rather than through a single estimated coefficient, and, on the much larger
 69 global sample, a country-fixed-effects specification that the primary sample’s
 70 single wave cannot support.

71 This paper contributes to the literature in four respects. First, it pro-
 72 vides what is, to our knowledge, the first direct empirical test of whether
 73 SBFN-style green banking policy adoption changes the firm-level relation-
 74 ship between finance and green investment, rather than only the aggregate
 75 volume of green lending reported by banks themselves—a firm-level, demand-
 76 side complement to the bank-level, supply-side evidence that dominates the
 77 existing green-credit-policy literature. Second, it extends the World Bank
 78 Enterprise Survey’s Green Economy module—used until now mostly within
 79 single countries or single regional waves—into a harmonized 41-country pri-
 80 mary sample spanning Europe, Central Asia, the Caucasus, and the Middle
 81 East and North Africa, giving the paper a breadth of institutional varia-
 82 tion that a single-country or single-region design cannot offer. Third, and
 83 unusually for this literature, we subject the primary sample’s central find-
 84 ing to an independently assembled second test: a 162-economy global sam-
 85 ple built from a World Bank standardized cross-country database spanning
 86 2021–2026 that, to our knowledge, no published study of green banking pol-
 87 icy has yet exploited, extending coverage for the first time into high-income
 88 economies entirely outside the SBFN’s own remit and allowing a country-
 89 fixed-effects specification the primary sample’s single-wave design cannot
 90 support. That the same qualitative result—a robust finance-green channel,
 91 no detectable institutional moderation—survives this independent replica-
 92 tion is, we think, considerably more informative than either sample could
 93 be alone. Fourth, methodologically, the paper demonstrates—with the same
 94 underlying question tested twice, on two different samples, under four differ-
 95 ent estimators—exactly where and why a classical fixed-effects approach fails

96 to detect (or, when country fixed effects are usable, correctly fails to find) an
97 institutionally-conditioned effect, offering applied researchers working with
98 similarly structured cross-country firm data a concrete illustration of when
99 the extra modelling investment is worth making and when a null result is
100 simply a null result.

101 The paper also speaks directly to a growing body of recent scholarship in
102 this journal on the institutional and financial determinants of firm and bank
103 behaviour in emerging and transition economies. Yan et al. (2024) study
104 how environmental protection taxes reshape green productivity among Chi-
105 nese listed firms; Barra and Falcone (2026) examine how institutional factors
106 condition environmental performance across a global sample of economies;
107 Heyert and Weill (2025) and Bach et al. (2025) use firm- and bank-level data
108 across dozens of countries to study, respectively, trust-driven financial inclu-
109 sion and credit constraints’ effect on corporate tax behaviour in Vietnamese
110 firms; Li et al. (2026) study how shadow-banking engagement reshapes trade-
111 credit financing among Chinese non-financial firms; and Horvath et al. (2025)
112 apply Bayesian model averaging to the cross-country determinants of finan-
113 cial development, a methodological precedent we build on directly in adopting
114 a Bayesian estimation strategy in Section 4. This paper sits at the intersec-
115 tion of these threads—institutions, green performance, and firm-level financ-
116 ing behaviour, studied jointly rather than pairwise—and, to our knowledge,
117 is the first among them to test whether a specific, dateable policy-adoption
118 event (SBFN membership) moderates the finance-green relationship at firm
119 level across a comparably broad cross-section of adopting economies.

120 The remainder of the paper is organized as follows. Section 2 sets out

121 the institutional background of the SBFN framework and the concepts and
122 trends in green banking policy adoption across our sample. Section 3 devel-
123 ops the paper’s three hypotheses from the state-capacity and institutional-
124 complementarity literature. Section 4 describes the primary and global sam-
125 ples, variable construction, and the empirical strategy for each. Section 5
126 presents baseline, hierarchical, and heterogeneous-effect results on the pri-
127 mary sample, followed by the independent global-sample replication. Sec-
128 tion 6 discusses the findings’ economic magnitude and their implications for
129 how green credit policy is designed and sequenced, and Section 7 concludes.

130 **2. Institutional background: green banking policy in emerging and** 131 **transition economies**

132 The Sustainable Banking and Finance Network (SBFN)—established in
133 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
134 expanded mandate covering both banking and non-bank financial regulation—
135 is the principal vehicle through which financial-sector regulators and banking
136 associations in emerging and developing economies have coordinated the de-
137 sign of national sustainable-finance policy. Hosted and technically supported
138 by the International Finance Corporation, the network now counts roughly
139 seventy member institutions (central banks, banking regulators, and banking
140 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
141 East Asia and the Pacific, Europe and Central Asia, Latin America and
142 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
143 national Finance Corporation / Sustainable Banking and Finance Network,
144 2025). Notably for our sample, the network’s membership is essentially dis-

145 joint from the European Union: not one of the sixteen EU member states in
146 our WBES sample appears on any SBFN membership roster we could locate,
147 consistent with the network’s explicit mandate to serve markets outside the
148 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
149 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
150 most of our survey waves in any case).

151 Membership itself, however, is only the network’s coarsest measure of
152 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
153 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
154 2023–2024) and country-level Progress Report addenda—scores each mem-
155 ber jurisdiction’s actual regulatory output against several policy pillars, most
156 centrally environmental-and-social risk management and, in later report vin-
157 tages, ESG disclosure integration and green-finance-flow measurement, and
158 places each jurisdiction on a progression from an initial “Commitment” or
159 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
160 ing” and, for the most mature frameworks, “Consolidating.” The variation
161 this produces across our sample is considerable and is itself part of the paper’s
162 identifying variation, not just a robustness afterthought. Mongolia’s Bank
163 of Mongolia was a founding SBFN member in 2012 and had already reached
164 the network’s “Advancing” implementation stage by the time of the country’s
165 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
166 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
167 member, had progressed by its 2022 Enterprise Survey wave into the net-
168 work’s ESG-integration pillar at an “Advancing” classification, while India’s
169 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained

170 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
 171 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
 172 had progressed only as far as an initial “Commitment” stage by their 2019
 173 survey waves. A further, and for identification purposes equally important,
 174 group of economies in our sample joined SBFN only after their Enterprise
 175 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
 176 istan (2023), and most of the Western Balkan and South Caucasus economies
 177 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
 178 their survey year, even though several are now members. This ex-ante cod-
 179 ing is deliberate: it is the policy environment a firm actually operated under
 180 at the moment its Green Economy module responses were recorded that is
 181 relevant to our test, not the policy environment that jurisdiction eventually
 182 adopted.

183 The economic logic behind the SBFN framework, and the reason it is a
 184 plausible channel for firm-level green investment rather than a purely sym-
 185 bolic commitment, rests on a straightforward supervisory mechanism: once
 186 a banking regulator formally adopts environmental-and-social risk manage-
 187 ment guidelines, commercial banks under its supervision face a compliance
 188 incentive to price environmental risk into individual lending decisions and,
 189 in many of the more advanced frameworks, to report a green-lending share
 190 to the supervisor directly. Existing evidence on China’s much-studied green
 191 credit guidelines documents exactly this channel operating at bank and firm
 192 level—reduced lending to high-polluting borrowers and improved environ-
 193 mental performance among firms that retain access to credit (Li et al., 2023;
 194 Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller liter-

195 ature on Bangladesh’s own long-running green-banking guidelines points in
196 the same direction (Khairunnessa et al., 2021). What neither literature has
197 established is whether this channel is detectable across a broad, heteroge-
198 neous cross-section of SBFN-style adopters at the firm level using a common
199 survey instrument, nor whether the strength of the channel depends on the
200 regulatory capacity available to enforce the underlying guideline—the two
201 questions this paper turns to next.

202 **3. Literature review and hypothesis development**

203 *3.1. Access to finance and firm-level green investment*

204 A firm’s decision to install pollution-abatement equipment, retrofit ma-
205 chinery for energy efficiency, or absorb the upfront cost of an emissions-
206 monitoring system is, first and foremost, an investment decision, and in-
207 vestment decisions are shaped by whether a firm can finance them. The
208 theoretical case is close to a standard credit-constraint story: green invest-
209 ments of the kind captured in the WBES Green Economy module (heat-
210 ing and cooling upgrades, on-site renewable generation, waste-minimization
211 retrofits, energy-management systems) typically carry a large upfront capi-
212 tal outlay against a payback stream realized only over several years, which is
213 exactly the profile of investment most sensitive to a firm’s access to external
214 finance rather than to retained earnings alone. Firm-level evidence bears
215 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
216 facturing firms, show that credit-constrained firms are measurably less likely
217 to adopt cleaner production practices even when they report an equivalent
218 willingness to comply with environmental regulation. De Haas et al. (2025)

document, across a large cross-country firm sample, that managerial and financial barriers operate as distinct but jointly binding constraints on firms' green transition investments, and Kalantzis et al. (2022) reach a closely related conclusion using EBRD transition-economy survey data, finding that firms' green investment is driven by a mix of financing conditions and climate exposure rather than by climate motivation alone—precisely the joint finance-and-institutions framing this paper tests directly. Costa et al. (2024) and Truong (2026) find, respectively in OECD and Vietnamese data, that financing constraints blunt firms' green investment response to environmental policy pressure that would otherwise induce it, while Aristei and Gallo (2023) show that firms with better access to credit are both more likely to have adopted green management practices before a shock and better able to sustain them through one. Ullah (2025) narrows this further to unlisted SMEs specifically, showing that financial constraints bind corporate environmental responsibility at smaller firm scales—a firm-size channel this paper's own causal-forest evidence (Section 5) also implicates, though, as we report there, without recovering a clean monotonic size gradient of its own. Siewers et al. (2024) extend the firm-level environmental-performance literature to a global-value-chain setting, showing that a firm's position in international production networks shapes its environmental performance net of domestic financing conditions, and Bambe et al. (2026) show that environmental policy stringency itself reshapes firm efficiency in developing economies, underscoring that the firm-level margin, not only the aggregate or bank-level one, is where much of this literature's true test now lies. Taken together, this literature converges on a simple first-order proposition, which any test of a

244 green-credit-policy channel must first establish before asking whether policy
245 moderates it:

246 **H1.** Firms with access to formal bank credit (a credit line or
247 loan from a financial institution) are more likely to have adopted
248 green production practices than otherwise-similar firms without
249 such access.

250 *3.2. Does green banking policy strengthen the finance-to-green-investment*
251 *channel?*

252 H1 describes a channel, not a policy. The entire premise of the SBFN
253 framework and its national analogues—and of the much larger literature on
254 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang
255 et al., 2023)—is that a supervisory mandate directing banks to price envi-
256 ronmental risk into lending, or to preferentially finance green activity, should
257 widen the gap in green-investment behaviour between firms with and with-
258 out bank access relative to a counterfactual world without the policy: banks
259 in adopting jurisdictions have a compliance incentive to seek out, and price
260 favourably, exactly the borrowers undertaking the kind of investment our
261 outcome variable measures, while credit-constrained firms in the same ju-
262 risdiction see no comparable improvement in their financing terms. This
263 is a moderation hypothesis, not a simple main-effect one, and it is worth
264 being explicit that most of the existing supporting evidence—again, over-
265 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
266 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
267 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
268 is drawn from single-country studies of a single, unusually well-resourced

269 green-credit regime. Whether the same interaction holds across the far more
270 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
271 work, many of them at a far earlier and less resourced stage of implementation
272 than China's, is untested. This motivates:

273 **H2.** The positive association between bank credit access and
274 green-practice adoption (H1) is stronger in economies that had,
275 as of the survey year, adopted an SBFN sustainable-finance policy
276 framework than in economies that had not.

277 *3.3. Institutional complementarity: does policy credibility depend on state*
278 *capacity?*

279 A green banking guideline is only as effective as the supervisory apparatus
280 behind it. This is the institutional-complementarity argument at the centre of
281 the comparative-institutions, law-and-finance, and state-capacity literatures.
282 Hall and Soskice (2001)'s varieties-of-capitalism framework is the founda-
283 tional statement of the underlying logic in its most general form: formally
284 similar policy instruments produce systematically different outcomes depend-
285 ing on how they interlock with the surrounding institutional architecture—
286 a regulatory mandate that functions as intended within one configuration
287 of complementary institutions may function quite differently, or not at all,
288 transplanted into another. Applied more narrowly to financial regulation
289 specifically, the same logic runs through the law-and-finance and state-capacity
290 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
291 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
292 2005, 2008) established that formally identical financial-contracting rules pro-
293 duce very different real-economy financing outcomes depending on the quality

294 of the legal and regulatory institutions that enforce them, while Acemoglu
295 et al. (2001) showed the same logic operating at the level of a country’s en-
296 tire institutional endowment. Applied to our setting, the argument is direct:
297 a bank supervisor’s environmental-risk guideline can only reshape lending
298 behaviour to the extent that the supervisor has the regulatory capacity—
299 examination authority, credible sanctions, reliable disclosure enforcement—
300 to make compliance with it more than nominal. Where that capacity is
301 weak, an SBFN framework may exist on paper (a formal “Commitment” or
302 “Formulating” stage classification, in the network’s own terminology) with-
303 out meaningfully altering bank lending behaviour on the ground, precisely
304 the state-capacity/policy-credibility distinction that motivates this special
305 issue. This is also the logic behind the small but growing strand of work on
306 institutions and environmental performance published in this journal specifi-
307 cally: Barra and Falcone (2026), studying a global sample of economies, find
308 that institutional-quality indicators condition environmental performance in
309 ways that are not reducible to income level alone, a result our design is built
310 to test at the firm rather than the country level. We proxy regulatory ca-
311 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
312 Quality estimate, matched to each economy’s survey year, and hypothesize
313 a three-way interaction:

314 **H3.** The policy-moderation effect described in H2 is itself in-
315 creasing in regulatory quality: SBFN adoption strengthens the
316 credit-to-green-investment channel most where the supervising
317 jurisdiction has the institutional capacity to make the underly-
318 ing guideline credible, and least—potentially not at all—where it

319 does not.

320 Hypotheses H2 and H3 are, by construction, statements about how a
321 country-level variable reshapes a firm-level relationship—precisely the two-
322 level structure that Section 4 argues a classical single-level fixed-effects re-
323 gression is poorly suited to recover with a country-cluster count in the low
324 forties, and that motivates the hierarchical and machine-learning estimators
325 we turn to after reporting that classical benchmark.

326 **4. Data and methodology**

327 *4.1. Data sources and sample construction*

328 The paper’s firm-level data come from the World Bank Enterprise Sur-
329 veys (WBES), a standardized, stratified-random-sample survey of formal-
330 sector, non-agricultural, non-extractive private firms with at least five em-
331 ployees, conducted by the World Bank’s Enterprise Analysis Unit. Our pri-
332 mary analysis sample is the World Bank’s own harmonized cross-country
333 file combining the 41 Europe and Central Asia (ECA) and Middle East and
334 North Africa (MENA) economies surveyed under a single, identical question-
335 naire between 2018 and 2020, the survey wave in which the Bank’s Green
336 Economy module—a battery of questions on emissions monitoring, energy-
337 consumption targets, and the adoption of specific climate-friendly production
338 measures—was fielded in full to every respondent. This yields 28,042 firm-
339 level observations across 41 economies (Table 2). We supplement this primary
340 sample with 15,556 additional firm observations from six further economies
341 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
342 Timor-Leste 2021) in which a shortened, partially randomized version of the

343 Green Economy module was fielded; because the exact item wording and ran-
344 domized sub-module assignment differ across these six economies, we do not
345 pool them into the main harmonized outcome variable, and instead use them
346 as an out-of-region external-validity check on a narrower, honestly-matched
347 pair of indicators (CO2-emissions monitoring, available in five of the six;
348 waste-minimization adoption, available in five of the six, with different spe-
349 cific economies covered by each), reported separately in Section 5.

350 The primary sample, while broad by the standards of the existing green-
351 credit-policy literature, is not the full universe of WBES economies carrying
352 *any* Green Economy content, and we build a second, independent sample
353 to exploit the rest of it. The World Bank also maintains a standardized
354 cross-country database (2006–2026) that harmonizes a common set of core
355 variables across every Formal Sector Enterprise Survey wave it has ever
356 fielded, 292,314 firm-level observations across 471 country-years. Within
357 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
358 into high-income economies entirely outside the SBFN’s remit such as Aus-
359 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
360 Economy item set: climate-damage exposure and, more relevantly for our
361 outcome construct, CO2-emissions monitoring, alongside the same standard
362 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
363 stacle) used in the primary sample. We supplement this with Bangladesh
364 2022 and Indonesia 2023 — two economies whose Green Economy responses
365 used an alternative randomized-module variable naming convention and so
366 are absent from the standardized database’s harmonized columns, but which
367 we can recover from our own extraction of the underlying country files —

368 for a combined global sample of 162 economies and 100,115 firms (89,797
369 with complete data on all covariates). Firm size, sector, and productivity
370 controls for this sample are drawn from the Bank’s companion Productivity
371 Database, merged by firm identifier. We refer to this as the *global sample*,
372 and to CO2-emissions monitoring as its outcome variable throughout; it is
373 the paper’s vehicle for testing whether the primary sample’s central find-
374 ing replicates independently, at far greater scale, and under an identification
375 strategy — country fixed effects — that the primary sample’s single-wave
376 design cannot support (Section 4).

377 Country-level data are drawn from two sources for both samples. For the
378 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
379 and progression-framework stage as of each economy’s survey year are com-
380 piled from the Sustainable Banking and Finance Network’s Global Progress
381 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
382 denda, and the network’s live country-profile data portal, cross-checked against
383 IFC press releases announcing individual member accessions; economies that
384 joined SBFN only after their WBES survey year are coded as non-adopters
385 as of that year (Section 2). For the global sample’s remaining 157 economy-
386 years, we draw membership status and join year directly from the SBFN data
387 portal’s full published member roster (67 members as of the portal’s most
388 recent update), applying the identical ex-ante rule: an economy is coded as
389 an adopter only if its join year precedes its survey year. Six economy-years
390 in the global sample (Central African Republic, Cameroon, Chad, Republic
391 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
392 the roster under a regional entry, “Central African States,” which we inter-

pret as referring to the CEMAC zone’s common banking supervisor; four small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St. Vincent and the Grenadines, all 2025) are similarly recorded under “Eastern Caribbean States,” which we interpret as the Eastern Caribbean Central Bank’s member states. Both interpretations are flagged explicitly here as inferences rather than explicit per-country listings. Regulatory capacity is measured with the World Bank’s Worldwide Governance Indicators (WGI) Regulatory Quality estimate, matched to each economy’s exact survey year (or, where not yet published, the most recent prior year available) via the World Bank DataBank API; WGI does not cover Taiwan, which is accordingly dropped from global-sample regressions.

4.2. Variable construction

Outcome. Our primary dependent variable, *green practice adoption*, is a binary indicator equal to one if a firm reports having adopted at least one of seven Green Economy module practices over the preceding three years: more climate-friendly on-site energy generation, an energy-management system, waste-minimization or recycling measures, air-pollution control measures, other pollution-control measures, any energy-efficiency measure, or the use of on-site renewable energy. We also construct a continuous *green adoption index*, the mean of the same seven binary items, for robustness.

Treatment. *Bank credit access* is a binary indicator equal to one if the firm reports holding a line of credit or loan from a formal financial institution at the time of the survey.

Country-level moderators. *SBFN member* is a binary indicator equal to one if the economy had adopted a national SBFN sustainable-finance

418 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
 419 ulatory Quality estimate for the survey year (continuous, standardized to
 420 mean zero, unit standard deviation within the estimation sample for the
 421 hierarchical and machine-learning specifications).

422 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
 423 own sampling strata), broad sector (manufacturing, services, construction),
 424 log sales, an exporter dummy (any direct or indirect export share), and a
 425 foreign-ownership dummy (any private foreign ownership share).

426 Table 1 lists every variable used in the analysis together with its exact
 427 WBES question code and source, in the interest of full transparency about
 428 variable construction given the multi-source nature of the merge.

429 4.3. Empirical strategy

430 We estimate the relationship between bank credit access and green prac-
 431 tice adoption, and its moderation by SBFN status and regulatory quality,
 432 in four stages of increasing methodological sophistication, each addressing a
 433 specific limitation of the one before it.

434 **Stage 1 — classical benchmark.** We first estimate a pooled logistic
 435 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

436 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a
 437 sector fixed effect, with standard errors clustered by country. We deliberately
 438 do not include country fixed effects in this specification: because SBFN_c is
 439 constant within country in a single cross-sectional wave, a country fixed effect

440 would absorb it—and its interaction with $\text{Credit}_{i,c}$ —completely, making β_2
 441 and β_3 unidentifiable by construction. This is not a modelling choice we
 442 are free to make differently; it is the reason a purely classical fixed-effects
 443 approach cannot answer H2/H3 as posed, and we report this specification,
 444 extended with a triple interaction for Regulatory quality_c, precisely to make
 445 that limitation visible.

446 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 447 estimate the same relationship allowing country to enter as a random rather
 448 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

449 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
 450 in PyMC. This specification is the paper’s primary vehicle for testing H2
 451 and H3: because country enters through a variance component rather than
 452 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
 453 remain identified, while the random slope $u_{1,c}$ still absorbs unobserved coun-
 454 try heterogeneity in the credit-green relationship and, together with partial
 455 pooling across countries, gives more defensible inference than cluster-robust
 456 standard errors computed over only 41 clusters (Cameron and Miller, 2015).

457 **Stage 3 — causal forest / double machine learning.** To avoid
 458 imposing a linear interaction functional form on how the treatment effect

of credit access varies across the institutional landscape, we additionally estimate a Causal Forest with Double Machine Learning (Wager and Athey, 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted nuisance models for the outcome and treatment propensity, and firm size, export status, SBFN status, and regulatory quality as effect-modifying features. This lets us recover conditional average treatment effects (CATEs)—the estimated effect of credit access on green adoption, separately for SBFN members versus non-members, and separately by regulatory-quality tercile—without imposing that the moderation take the single-coefficient interaction form assumed in Stages 1–2.

Stage 4 — small-sample supplementary check. We additionally re-estimate a reduced-form version of the credit-green relationship using a waste-minimization-adoption outcome, available in four economies (India 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy modules asked this item under directly comparable wording. This outcome is not part of the standardized cross-country database underlying Stage 5 below, so it offers genuinely independent, if small-sample, supplementary evidence.

Stage 5 — global-sample replication with country fixed effects. Finally, and centrally, we re-estimate the full specification on the 162-economy global sample described in Section 4, using CO2-emissions monitoring as the outcome. Because this sample spans far more country clusters than the primary sample, we can now afford a country-fixed-effects specification: δ_c absorbs SBFN_c and RegQuality_c’s main effects (collinear with the fixed effect by construction, since both are constant within country-year) while leaving

484 their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access
485 itself varies within country. This is a materially sharper test of H2/H3 than
486 Stage 1’s clustered-SE approach, on an entirely independent sample and out-
487 come from Stages 1–3; we report it alongside a no-fixed-effects specification
488 for direct comparability with the primary sample’s Table 4.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Mod- ule
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_ quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufactur- ing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $>0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $>0\%$	D3B, D3C	WBES Sales Module

489 5. Results

490 5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

491 Table 2 reports the full 41-economy sample composition; firm counts range
492 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
493 practice-adoption rate ranges more than three-fold across economies, from
494 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
495 SBFN policy status: 8 of the 41 primary-sample economies had adopted
496 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
497 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
498 economy sample once the five (of six) SBFN-member small-sample supple-
499 mentary economies are counted—and every European Union member state
500 in the sample is, consistent with SBFN’s mandate, a non-adopter. At the
501 firm level, bank credit access correlates positively with the green-adoption
502 index ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived
503 finance obstacle ($r = -0.02$)—a first hint, well before any regression, that re-
504 alized access to credit rather than perceived financing difficulty is what tracks
505 green behaviour, and the two are evidently not interchangeable measures of
506 the same underlying constraint.

507 Figure 2 previews the paper’s central finding graphically before a sin-
508 gle regression is estimated: plotting each economy’s green-adoption rate

509 against its regulatory-quality score and fitting separate lines for SBFN mem-
 510 bers and non-members produces two lines of very similar, modestly positive
 511 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 512 member economies show systematically *lower* raw green-adoption rates at
 513 every level of regulatory quality, not a steeper relationship between regu-
 514 latory quality and adoption. That combination—a level difference with no
 515 slope difference—is exactly the signature the regression evidence below con-
 516 firms formally.

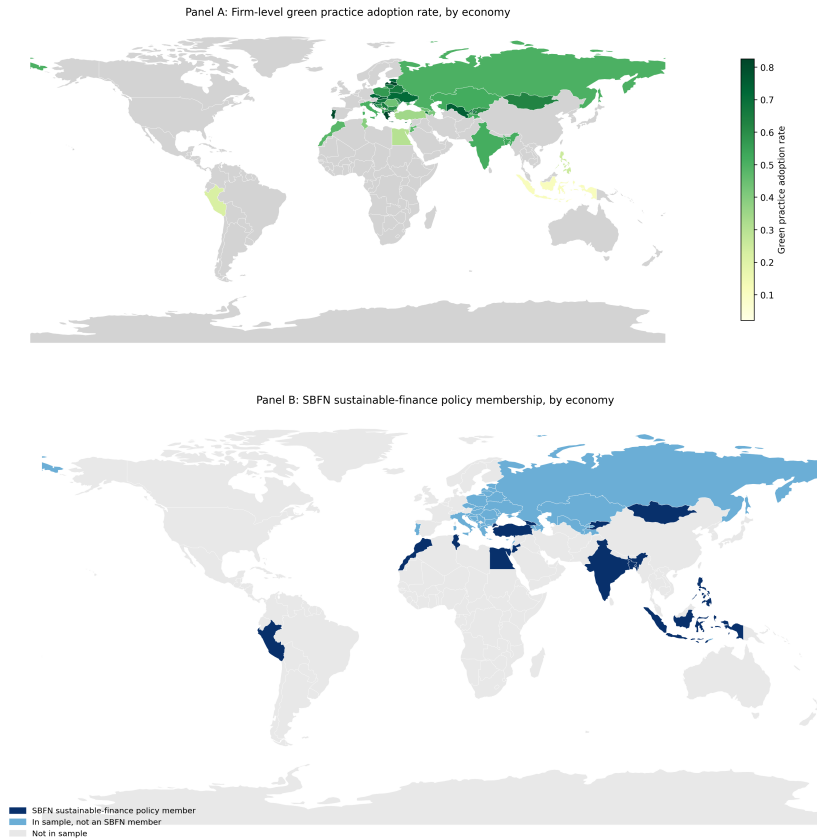


Figure 1: Green practice adoption and SBFN policy membership, by economy.

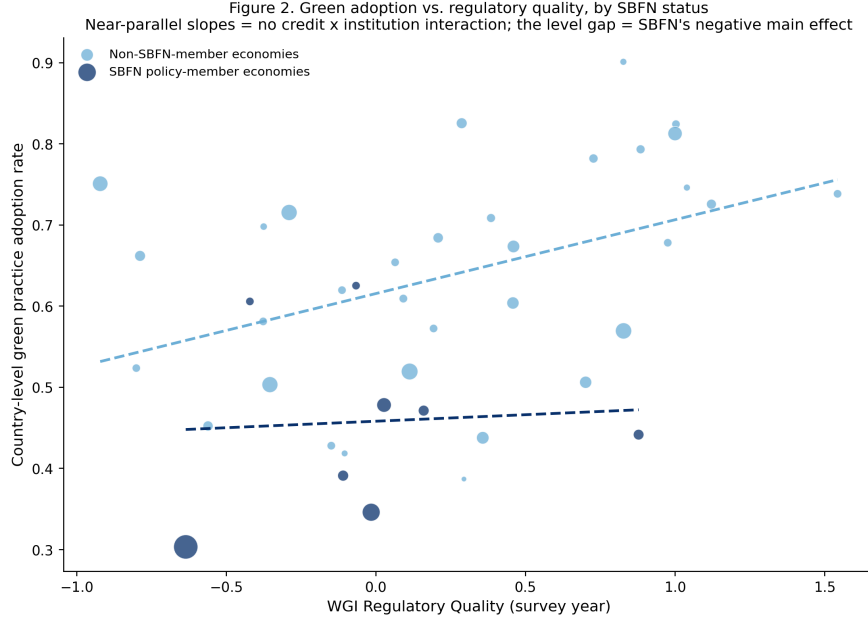


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3's fully saturated triple interaction with regulatory quality tells the same story

at greater length: every interaction term involving credit access is small and insignificant, while regulatory quality’s uninteracted relationship with adoption is positive, if only marginally significant in this particular fully-saturated specification ($b = 0.278$, $p = 0.110$).

As flagged in Section 4, this specification cannot include country fixed effects without mechanically absorbing the very SBFN and regulatory-quality terms the hypotheses concern, so what Table 3 shows is a genuine test of H2/H3, not a placeholder—and that test returns no interaction. The question the remaining two stages are built to answer is whether that null result reflects a real absence of policy moderation, or an artefact of forcing a two-level question through a single-level model with country-clustered standard errors computed over only 41 clusters, a setting in which such standard errors are known to be unreliable (Cameron and Miller, 2015).

5.3. Bayesian hierarchical model — primary test of H2/H3

The hierarchical specification—country-varying random slopes on credit access, cross-level interactions with SBFN status and standardized regulatory quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and ess_{bulk} between 797 and 2,270—is built specifically to give H2 and H3 a fair hearing that the absorbed classical specification could not. It does not change the conclusion. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval $[0.18, 0.49]$), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN mem-

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

bership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2 and H3, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we can state that SBFN status and regulatory quality shift the *level* of green adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

568 5.4. Causal forest / heterogeneous treatment effects

569 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
570 pose no linear-interaction functional form and let the treatment effect vary
571 flexibly across firm and country covariates. The overall average treatment
572 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
573 tent with, though smaller in magnitude and less precisely estimated than,
574 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more
575 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
576 ric estimator relative to a parametric logit on the same data. Critically,
577 the conditional average treatment effects triangulate with Section 5.3 rather
578 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
579 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /
580 0.075 across the low/middle/high regulatory-quality terciles respectively—
581 three different institutional cuts, three essentially flat effects, all visibly over-
582 lapping in Figure 5. Two entirely independent estimators, one parametric
583 and hierarchical, one nonparametric and forest-based, agree on the absence
584 of institutional moderation.

585 The forest’s feature-importance decomposition adds a genuinely new re-
586 sult rather than only confirming the null: of the four candidate effect-modifiers,
587 firm size (log sales) accounts for 75.6% of the model’s explained heterogene-
588 ity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9%
589 for exporter status. Whatever heterogeneity exists in how strongly credit ac-
590 cess predicts green adoption is, on this evidence, primarily a firm-level rather
591 than a country-institutional phenomenon. Collapsing this into the Enterprise
592 Surveys’ own coarse size strata, however, does not reveal a clean monotonic

593 gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms,
594 and 0.057 for large firms (Table 6; all with wide, overlapping confidence in-
595 tervals spanning zero), which if anything runs opposite to a simple “larger
596 firms better absorb a green investment’s fixed costs” story rather than con-
597 firming it. Read alongside the 75.6% importance share for the continuous
598 size measure, we take this as evidence that the heterogeneity the forest is
599 detecting is more granular or non-monotonic across the firm-size distribu-
600 tion than a three-category collapse can reveal, rather than evidence for any
601 particular direction of a size gradient—a qualification we report explicitly
602 rather than paper over with a tidier-sounding but unsupported claim. This
603 redirects, rather than closes, the paper’s institutional-moderation question,
604 and we return to it in Section 6.

605 *5.5. Small-sample supplementary check: waste-minimization adoption*

606 Before turning to the global-sample replication that is this paper’s second
607 major test (Section 5.7), Table 7 reports a smaller supplementary check using
608 waste-minimization adoption — an outcome not captured in the standardized
609 cross-country database underlying the global sample, and so a genuinely in-
610 dependent (if small-sample) data point — available in four economies whose
611 Green Economy modules asked this item under directly comparable word-
612 ing: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of
613 these four are themselves SBFN members at various progression stages, so
614 this sample cannot re-test the SBFN-adoption contrast; what it can test is
615 whether the credit-to-green channel documented in Sections 5.2–5.4 is a pe-
616 culiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes:
617 pooled across the four economies, having an overdraft facility is associated

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701

Feature	Importance
SBFN member	0.0257
Regulatory quality	0.1993
Log sales	0.7562
Exporter dummy	0.0189

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

with significantly higher odds of waste-minimization adoption ($b = 0.422$,
 $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green
relationship replicates on a different continent, a different half-decade, and
a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a
preview, on four economies, of the much larger replication in Section 5.7.

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

5.6. Additional robustness

Table 8 subjects the primary-sample finding to four further checks, all
retaining sector fixed effects, firm controls, and country-clustered standard
errors (no country fixed effects, for the reason given in Section 4). First,
replacing the binary outcome with the continuous seven-item green-adoption
index and estimating by OLS reproduces the same pattern to the decimal:
credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main
effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistin-
guishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance mea-
sures produces one genuine point of nuance worth reporting rather than
smoothing over. Using overdraft-facility access instead of a credit line/loan,
the main overdraft effect remains positive and significant ($b = 0.274$, $p =$

0.011), the SBFN main effect remains negative and significant ($b = -0.905$,
 $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and sig-
nificant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper
that clears a conventional significance threshold, and it runs opposite to the
direction H2 predicts. Using the reverse-coded perceived finance-obstacle
measure in place of realized access, by contrast, both the main effect and
its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interac-
tion $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation
noted in Section 5.1. We read the overdraft result as suggestive rather than
conclusive given it is one significant coefficient among more than a dozen
interaction tests across the paper, but it is at least directionally consistent
with a substantive story worth flagging for future work: if SBFN-guided
lending is channelled preferentially through the term loans and credit lines
that finance long-horizon capital expenditure, rather than through short-term
overdraft facilities, we might expect exactly this pattern—a null-to-positive
interaction on credit lines and a negative one on overdrafts—though our
data cannot directly test whether SBFN-guided credit is disproportionately
term-structured.

Third, splitting the sample into manufacturing ($N = 13,340$) and services
($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main
effect, and the null credit $\times$ SBFN interaction all replicate within each sector
separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b =$
 -0.040 , $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b =$
 0.195 , $p = 0.426$)—the paper’s central null is not an artefact of pooling two
structurally different production technologies.

Table 8: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

5.7. Global-sample replication with country fixed effects

The preceding five subsections establish the paper’s central finding on the primary sample. We now subject that finding to what we consider the most demanding test available to us: complete re-estimation on an independently assembled, 162-economy global sample (Section 4), using a different outcome variable (CO2-emissions monitoring rather than the seven-item composite), different survey years (2021–2026 rather than 2018–2020), and, for the first time in the paper, a country-fixed-effects specification that this much larger set of clusters can support. Figure 3 maps the combined reach of the two samples together: 166 distinct economies, spanning every region and, through the global sample specifically, extending for the first time in this literature into high-income economies with no SBFN involvement at all.

Table 9 reports two specifications. Column M1, without country fixed effects, is the direct global-sample analogue of Table 4’s primary-sample baseline: credit access remains positive and significant ($b = 0.198$, $p = 0.039$, $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$, $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and every regulatory-quality interaction are statistically indistinguishable from

679 zero. Column M2 imposes country fixed effects, absorbing the SBFN and
 680 regulatory-quality main effects by construction (they are collinear with δ_c ,
 681 since both are constant within a given country-year) while leaving their firm-
 682 level interactions with credit access fully identified, because credit access
 683 itself varies within country. This is a sharper test than anything the primary
 684 sample’s single 2018–2020 wave can support, and it returns the same answer
 685 with even more precision: credit access is strongly significant ($b = 0.249$,
 686 $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
 687 the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
 688 null.

689 We regard this as the paper’s most important robustness result. It is not
 690 a re-analysis of the same data under a different label: it is a different outcome
 691 construct, a different set of economies (only a handful of which overlap with
 692 the primary sample), a different half-decade, and an econometric specification
 693 — country fixed effects — that is only available at all because this second
 694 sample has enough clusters to support it. That the qualitative conclusion
 695 is identical across both samples and across the full set of four estimators
 696 reported in this paper (classical logit, Bayesian hierarchical, causal forest, and
 697 now country-fixed-effects logit) is, we think, considerably stronger evidence
 698 for the paper’s central claim than either sample could offer alone.

699 *5.8. Results summary across all specifications*

700 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
 701 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
 702 into a single view. Panel A shows the credit-access effect (H1): every point
 703 estimate is positive and every confidence interval excludes zero, in both sam-

Table 9: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

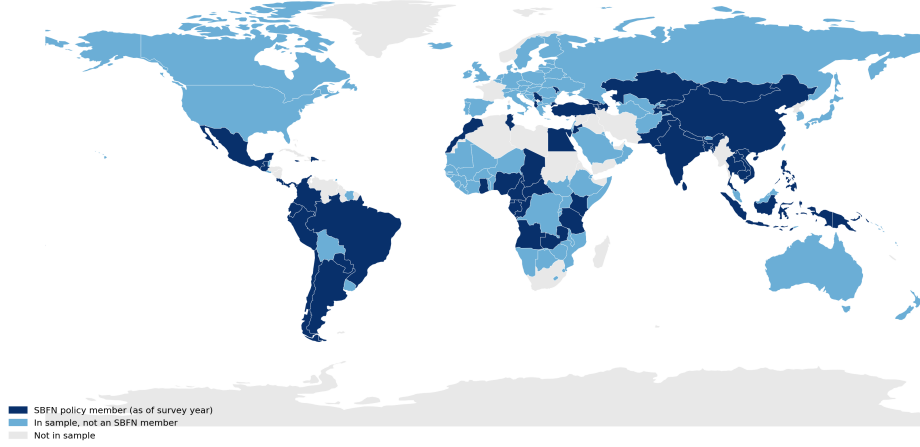


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

704 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
705 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
706 terval, across both samples and every estimator, straddles zero. Figure 5
707 shows the same non-result from a completely different angle—the causal
708 forest’s conditional average treatment effects broken out by SBFN status,
709 regulatory-quality tercile, and firm-size category all overlap both each other
710 and the overall average treatment effect. No subgroup split, by any variable
711 examined in this paper, recovers a distinguishable effect.

712 6. Discussion

713 Put in plain economic terms, the paper’s central finding is this: a firm
714 holding a bank credit line is roughly 13 percentage points more likely to have
715 adopted a green production practice than an otherwise identical firm without

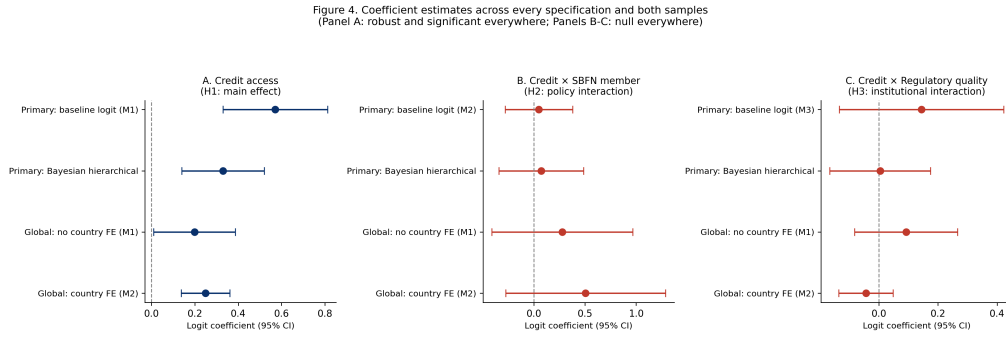


Figure 4: Coefficient estimates across every specification and both samples.

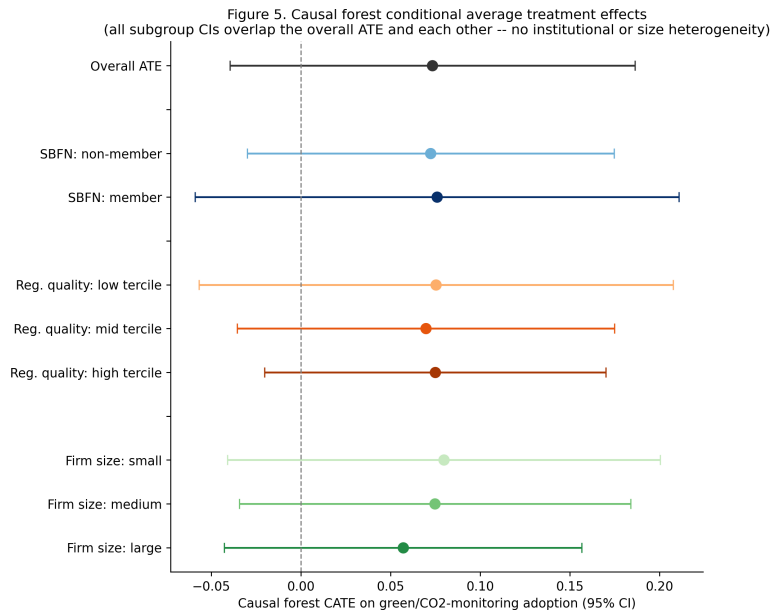


Figure 5: Causal forest conditional average treatment effects, by subgroup.

one—comparable in magnitude to the gap between a small firm and a large one—and that gap is essentially the same size whether the firm sits in an SBFN sustainable-finance policy adopter or a non-adopter, and essentially the same size whether the country’s regulatory quality sits in the bottom, middle, or top third of the sample (Figure 4). Firms in SBFN-adopting economies are, on average, some 15–20 percentage points *less* likely to have adopted a green practice at all—a level gap, not a slope gap.

This convergence matters most because of how differently the two samples could have failed to agree. The primary and global samples share almost no economies in common, were fielded in different half-decades, measure green practice adoption with entirely different survey items, and reach institutional settings—high-income, non-SBFN economies such as Australia and Germany—that the primary sample cannot touch at all. A finding that held only in the 41-country ECA-MENA sample could plausibly be a regional or period-specific artefact; a finding that survives unchanged when re-estimated on 162 largely non-overlapping economies under a sharper identification strategy is considerably harder to explain away as such.

This is worth dwelling on precisely because it complicates rather than confirms the implicit premise of much of the existing green-credit-policy literature, which is disproportionately built on evidence from China’s unusually mature, unusually well-resourced green credit guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025). Our sample includes economies at every stage of the SBFN’s own progression framework—from Jordan and the Kyrgyz Republic’s initial “Commitment” stage to Mongolia’s “Advancing” implementation a full seven years after founding membership—and across that

entire range, we find no evidence that formal policy adoption changes how a firm’s credit access translates into green investment behaviour. Section 5.4’s most novel result gives one plausible redirection of the question, though not a tidy answer to it: the heterogeneity that *does* exist in the credit-green relationship is overwhelmingly a firm-size phenomenon (75.6% of explained variance) rather than a country-institutional one (regulatory quality 19.9%, SBFN status a mere 2.6%)—yet, as Section 5.4 reports honestly, collapsing this into the Surveys’ own coarse Small/Medium/Large strata does not recover a clean gradient in either direction, so we cannot yet say whether it is small or large firms for whom the credit-green channel runs stronger, only that firm scale—in some more granular or non-monotonic form than a three-category split reveals—is where the forest locates most of the effect heterogeneity a country-level policy variable does not. What this does support is a reorientation of where policy attention belongs: if the operative constraint sits at the level of which individual firms can absorb a green investment’s fixed costs once financed, rather than at the level of whether a supervisory guideline nudges banks to lend more broadly, then a country-level supervisory mandate operating on the extensive margin of bank lending decisions may be aimed at the wrong margin regardless of its own institutional credibility—a question future work with a finer-grained firm-size or capacity measure is better placed to resolve than the coarse categories available here.

The one interaction that does clear a conventional significance threshold—a *negative* credit $\times$ SBFN effect when finance access is measured through overdraft facilities rather than term credit lines (Section 5)—is, we think, more consistent with this redirection than with a simple failure of green

766 banking policy. Overdrafts finance working capital, not the kind of multi-year
767 capital expenditure our outcome variable measures; if SBFN-guided lending
768 is disproportionately channelled through the term-loan instruments better
769 suited to financing a solar installation or an energy-efficiency retrofit, exactly
770 the pattern we find—a null-to-positive effect on credit lines, a negative one
771 on overdrafts—is what a supervisor successfully steering credit toward long-
772 horizon green investment, rather than short-horizon working capital, would
773 produce. We flag this as suggestive rather than established, since it is a
774 single significant coefficient among the many interaction tests reported across
775 Tables 3–8, and our data cannot directly verify the underlying loan-tenor
776 mechanism.

777 Read against this special issue’s organizing themes of state capacity and
778 policy credibility, our results sit closer to a state-capacity-is-necessary-but-
779 firm-level-frictions-bind-independently story than to a straightforward institutional-
780 complementarity one in which stronger regulatory quality unlocks a latent
781 policy effect that would otherwise lie dormant. Regulatory quality matters—
782 it raises the general baseline propensity to adopt green practices regardless of
783 a firm’s own financing arrangement—but it does not appear to be the miss-
784 ing ingredient that would let a green banking guideline reshape the credit-
785 to-green channel specifically. This suggests, tentatively, that the binding
786 constraint on green credit policy’s effectiveness in our sample is less about
787 whether the supervising jurisdiction has the state capacity to make the guide-
788 line credible, and more about whether firms below a certain scale can absorb
789 a green investment’s fixed costs at all once credit is available—a question
790 about firm-level absorptive capacity that sits one level down from the state-

791 capacity question the SBFN framework itself is designed to address.

792 **7. Conclusion**

793 This paper set out to test whether the wave of green banking policy
794 adoption coordinated through the Sustainable Banking and Finance Network
795 changes the relationship between firm-level access to finance and green invest-
796 ment behaviour, and whether any such moderation depends on the regulatory
797 capacity available to make the underlying policy credible. Using harmonized
798 World Bank Enterprise Survey data spanning 41 economies’ full 2018–2020
799 Green Economy Module rollout, a four-country waste-minimization supple-
800 mentary check, and an independently assembled 162-economy global sample
801 spanning 2021–2026 and reaching for the first time into high-income, non-
802 SBFN economies, we apply a deliberately escalating sequence of estimators—
803 a classical benchmark chosen to expose its own limitation, a Bayesian hier-
804 archical model built to give the moderation hypothesis a fair test, a causal
805 forest imposing no functional-form assumption at all, and, on the global
806 sample specifically, a country-fixed-effects specification the primary sample’s
807 single wave cannot support—and find four consistent results. First, access
808 to bank credit is robustly associated with a firm’s green practice adoption,
809 a relationship that survives every specification, subsample, and alternative-
810 outcome check in the paper and replicates independently in both the small
811 waste-minimization sample and the full 162-economy global sample. Second,
812 neither SBFN policy adoption nor regulatory quality moderates that rela-
813 tionship in any of the four estimators, on either sample, though both shift
814 the general level of green adoption directly and substantially in the primary

815 sample. Third, where genuine effect heterogeneity does exist, it is concen-
816 trated at the level of firm size rather than country institutions. Fourth, and
817 most importantly for the paper’s overall credibility, this entire pattern repli-
818 cates on a sample sharing almost no economies with the primary one, fielded
819 years later, measured with a different outcome item, and identified under a
820 materially sharper econometric strategy—the single strongest piece of evi-
821 dence in the paper that the null on institutional moderation is a real feature
822 of the data rather than an artefact of any one sample, period, or estimator.

823 For policy, the implication is not that green banking guidelines are ineef-
824 fective, but that our evidence does not support treating them as an effective
825 lever specifically for widening the gap in green investment between financed
826 and unfinanced firms—the margin most green-credit-policy literature implic-
827 itly assumes they operate on. If our tentative loan-tenor interpretation of
828 the overdraft-interaction finding is correct, policymakers designing or refining
829 SBFN-style frameworks may find more traction focusing supervisory atten-
830 tion on the term-lending instruments that actually finance green capital ex-
831 penditure, and on complementary measures—technical assistance, matching
832 grants, or credit guarantees—that address smaller firms’ absorptive-capacity
833 constraints directly, rather than assuming that a bank-level supervisory man-
834 date alone will reach firms regardless of scale.

835 We close with five limitations that bound how far these conclusions travel.
836 First, both samples are single cross-sections per country (Section 2 discusses
837 why a genuine panel on this outcome does not exist in the published WBES
838 data): they cannot rule out reverse causality (firms already predisposed to
839 green investment may find it easier to obtain credit) or unobserved, time-

840 invariant country characteristics correlated with both SBFN adoption and
 841 the baseline propensity to adopt green practices, and our identification rests
 842 on firm-level variation in credit access interacting with a plausibly firm-
 843 exogenous country-level policy indicator, not a natural experiment; the global
 844 sample's country fixed effects sharpen this only for the interaction terms, not
 845 for the credit main effect itself. Second, SBFN membership is a binary proxy
 846 for what the network's own multi-tiered progression framework treats as a
 847 genuinely graded process; a natural extension would model policy stage—
 848 Commitment through Consolidating—as an ordinal or continuous treatment
 849 rather than collapsing it to membership status, which our binary coding nec-
 850 essarily does, and which for the global sample's regional entries (the CEMAC
 851 and Eastern Caribbean groupings, Section 4) additionally rests on an infer-
 852 ence about which specific economies a regional listing covers rather than an
 853 explicit per-country statement. Third, the four-economy waste-minimization
 854 check cannot re-test the SBFN contrast itself, since three of its four economies
 855 are SBFN members, and combines non-identical outcome items across coun-
 856 tries by construction of the WBES's own shortened, randomized module de-
 857 sign in later survey waves—a genuine data-availability constraint rather than
 858 a methodological choice. Fourth, the global sample's CO2-monitoring out-
 859 come is a single item rather than the primary sample's seven-item composite,
 860 and is a narrower construct than "green practice adoption" in the sense the
 861 primary sample measures it; that the same qualitative pattern holds on this
 862 narrower measure is reassuring but does not establish that it would hold on
 863 a richer one, were a richer one available at global scale. Fifth, the World-
 864 wide Governance Indicators' regulatory-quality measure is a broad, economy-

865 wide governance indicator rather than a banking-supervision-specific capac-
866 ity measure; a supervisor-specific index, were one available consistently across
867 this country set, would sharpen the institutional-complementarity test con-
868 siderably. We leave each of these to future work.

869 **Declaration of generative AI and AI-assisted technologies in the** 870 **manuscript preparation process**

871 During the preparation of this work the author(s) used Claude (An-
872 thropic) to assist with data extraction and cleaning from World Bank Enter-
873 prise Surveys microdata (including identification and extraction of the stan-
874 dardized cross-country database underlying the global sample), construction
875 of the analysis datasets and merge with country-level SBFN and Worldwide
876 Governance Indicators data, execution of the statistical and machine-learning
877 analyses reported in Section 5, generation of Figures 1–3, and drafting of por-
878 tions of the manuscript text. After using this tool, the author(s) reviewed
879 and edited the content as needed and take full responsibility for the content
880 of the published article.

881 **Appendix A. SBFN policy status and regulatory quality, full sam-** 882 **ple**

883 Table A.10 reports, for all 47 economy-years in the paper’s primary and
884 small-sample supplementary check, the SBFN membership status coded for
885 the analysis, the join year where applicable, the SBFN Multi-Tiered Progress
886 Framework stage in force nearest the survey year, and the exact year of the
887 Worldwide Governance Indicators observation used. Full per-country source

888 citations (the specific SBFN Global Progress Report, country addendum, or
889 accession announcement consulted for every determination) are available in
890 the online supplementary material. The analogous 162-economy table for
891 the global sample (Section 4) is summarized by region in Appendix C and
892 provided in full in the online supplementary material and the replication
893 package.

Table A.10: SBFN policy status and regulatory quality, full 47-economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integra- tion)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Prepara- tion)	2020
Estonia 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Georgia 2019	1	2017	Developing (Implementation)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019
India 2022	1	2016	Developing (Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)	2023
Italy 2019	0	—	None	2019
Jordan 2019	1	2016	Commitment (Preparation)	2019
Kazakhstan 2019	0	—	None	2019
Kosovo 2019	0	—	None	2019
Kyrgyz Republic 2019	1	2018	Commitment (Preparation)	2019
Latvia 2019	0	—	None	2019
Lebanon 2019	0	—	None	2019
Lithuania 2019	0	—	None	2019
Malta 2019	0	—	None	2019
Moldova 2019	0	—	None	2019
Mongolia 2019	1	2012	Advancing (Implementation)	2019
Montenegro 2019	0	—	None	2019
Morocco 2019	1	2014	Advancing (Implementation)	2019
North Macedonia 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Peru 2023	1	2013	Advancing (ESG Integra- tion)	2023
Philippines 2023	1	2013	Advancing (ESG Integra- tion)	2023
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Repub- lic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementa- tion)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

894 **Appendix B. Bayesian model convergence diagnostics**

895 Convergence diagnostics for every focal parameter of the hierarchical
896 model reported in Table 4 are included directly in that table ($\hat{R}$ column); the
897 full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member)
898 to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional
899 minimum threshold of 400 effective samples per parameter for stable poste-
900 rior summaries at four chains. No divergent transitions were flagged during
901 sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup
902 draws each after 1,000 tuning iterations).

903 **Appendix C. Global-sample SBFN status, regional summary**

904 Of the 162 global-sample economy-years, 61 are coded as SBFN members
905 as of their survey year and 101 as non-members. Table C.11 reports the
906 regional distribution of the 61 members, drawn from the SBFN data por-
907 tal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste,
908 Bangladesh, and Indonesia) carry the deeper per-country sourcing already
909 reported in Appendix A, having been determined during the primary sam-
910 ple’s original research pass rather than from the portal roster alone.

911 **Additional material**

912 *Online appendix*

913 Appendix A, Appendix B, and Appendix C appear above; per-country
914 source citations at the level of detail compiled during data construction (in-
915 dividual SBFN Global Progress Report page references, country-addendum

Table C.11: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

916 URLs, and IFC accession-announcement links for every non-obvious deter-
917 mination in the primary sample; the full 162-row global-sample SBFN/WGI
918 table underlying Appendix C) are available in the online supplementary ma-
919 terial accompanying this submission.

920 *Data availability*

921 Data in support of the findings of this study is available from the cor-
922 responding author upon reasonable request, subject to the World Bank En-
923 terprise Surveys' own terms of use for the underlying firm-level microdata
924 (enterprisesurveys.org, free registration required), including the standard-
925 ized cross-country database and Productivity Database underlying the global
926 sample. The country-level SBFN policy-status (including the SBFN data
927 portal's member roster underlying the global sample) and Worldwide Gover-
928 nance Indicators datasets constructed for this paper, together with the full
929 analysis and replication code, are available in a public repository; the link is

930 withheld from this anonymized manuscript to preserve double-anonymized
931 review and will be provided to the editor upon submission and to readers
932 upon acceptance.

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